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Towards an Integrated Methodology for Assessing Items in Mathematics Assessments

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Abstract: There is no singular methodology for analyzing educational assessments. Moreover, educational assessment primarily serves to assess students over some curriculum rather than as collections of items with the most predictive power. It is thus necessary to develop a practical methodology to provide a robust analysis of an educational assessment that attends to both assessment of knowledge and predictive power. This study presents a unifying methodology that considers measures of validity, reliability, and knowledge to provide a robust analysis of an educational assessment. This paper presents classification bounds that inform labeling items as sufficiently discriminating for the purposes of an educational assessment. It then presents an analysis of 4 multiple-choice exams to illustrate how the metrics can be used in concert to identify validity, reliability, and knowledge at the item and assessment level. The paper concludes with a brief discussion of how this methodology can be applied to different types of assessments.

Keywords: *assessment, classical test theory, item response theory, distractor analysis*

Introduction

To assess an assessment, one must consider whether the assessment produces valid and reliable results according to its intended goals. Within assessment, validity refers to the extent a test measures what it purports to measure and reliability refers to the extent results can be replicated (Wu et al., 2016). For example, the goal of a placement exam is to classify students into the highest course they could take with success. Measuring the validity of a placement test would include analyzing the course grades for students placed and determining the impact of the placement test. Measuring the reliability of a placement test would include checking that students are consistently classified into some course. In contrast, the goal of a standardized test like the SAT is to measure some academic ability and report it in an ordered form to distinguish between high-performing and low-performing students. Measuring the validity of a standardized test may include checking the validity of individual questions (to ensure answering the question correctly relates to the student having some knowledge) while measuring reliability may include comparing correct response rates across versions within the same year and between different years.

The divide between classroom assessment and large-scale assessment further complicates the ways to measure validity and reliability (Heritage & Kingston, 2019). Important to classroom assessment is an underlying model of learning that forms a coherent sequence of curriculum. Classroom assessment is then meant to determine where a student is in this model of learning and specifically target their needs. It includes the need for actionable insight into student thinking as well as numerous questions closely aligned to Learning Objectives (Suurtamm et al., 2016). This knowledge-centric focus makes traditional measures for reliability and validity difficult to implement (Briggs, 2017). On the other side, large-scale assessments have well-structured methods for measuring reliability and validity given the copious amounts of data and focus on measuring the sum total of some knowledge in an infrequent fashion. While measuring knowledge gains has become more frequent in large-scale assessment with the use of learning maps (Kingston et al., 2016), it largely relies on providing a comparative score across some populations such as through descriptive statistics or general item response theory models (von Davier et al., 2019). While it may include questions aligned to Learning Objectives as seen in classroom assessments, special care is taken to ensure minimal overlap in questions to keep the size of the assessment as small as possible. The goal of this paper is to apply some measures of reliability and validity from large-scale assessments to the classroom in a way that is easy for an instructor to implement and utilize.

Literature Review

The overall goal of the proposed theoretical framework is to identify a collection of item measures centered around validity, reliability, and knowledge that can classify classroom assessments for practical use by instructors. This requires negotiation between psychometric measures for best predictiveness of items and common educational practices such as needing an item per learning objective regardless of how predictive answering a specific item is towards the final assessment score. Thus, this section walks through measures and classifies each measure based on published results from both psychometrics and small-scale educational assessment research.

Measures from Classical Test Theory

Classical Test Theory (CTT) is the analysis of assessments based primarily on students' overall (observed) assessment scores (Traub, 1997). In general, the core assumption underlying CTT is that a student's observed score is the sum of their "true score," which measures their actual understanding of an underlying concept, and an "error score," which can result from numerous sources. The true score is defined as the expectation of the observed score, meaning that "on average" the error will cancel out of the equation. CTT is commonly used by professional educators to validate an administered exam and is included when administering assessments using online learning management systems (Brown, 2023). For example, Canvas provides CTT measures such as mean score, standard deviation, item difficulty, and discrimination index for quizzes built within the Learning Management System (The Instructure Community, 2025). There are a few key statistics that can be measured when studying assessments using this framework: item difficulty and point-biserial correlation. A brief description for each of these key statistics follows.

Item Difficulty

For dichotomous items (correct/incorrect), item difficulty refers to the proportion of students that answered an item correctly. For polytomous items, responses may be weighed for partial credit. This

paper only considers dichotomous item difficulty. Item difficulty represents the mean score on an item. Its role in discrimination is more of an indicator that an item will not be able to be used to discriminate between high and low performing students than as an indicator of performance. In light of this, Lord (1955) suggests ideal item difficulties as slightly higher than halfway between a random guess and 100%, with 74% and 70% for four and five option multiple choice assessment items respectively. Note that while it is common for educational assessments to produce non-normal distributions (Ho & Yu, 2015) that will impact the normality assumption of Lord's ideal item difficulties, item difficulty will be one of many quantitative features used to classify assessment items.

It is common for educational research papers to only mention that average item difficulties are similar to those reported in other studies rather than providing strict boundaries. For example, Rush, Rakin, and White (2016) state "The mean ($\pm$ SE) percent correct ($83.3\% \pm 17.5$) was consistent with target values in professional education" (p. 7).

Point-biserial correlation

Point-biserial correlation (PBC) is an item-level statistic that measures the correlation between students' performance on an item and their performance on the exam as a whole. In other words, PBC describes how well the item discriminated between high-performing and low-performing students and is entirely distinct from item difficulty. Items with negative PBC or a PBC of zero are poor assessment items as they indicate that the item is answered incorrectly by students that perform well on the exam (suggests an item has an error) or that the item bears no relationship to student performance and can be removed without consequence. Categorization of PBCs for educational assessment purposes is not uniform, as assessments primarily serve to assess the curriculum rather than as collections of items with the most predictive power (Wu et al., 2016). When PBC is used to assess items, static recommended bounds are presented without rigorous justification. For example, Varma (2006) states "A point-biserial value of at least 0.15 is recommended, though our experience has shown that "good" items have point-biserials above 0.2" (p. 6), McGahee and Ball (2009) state "As a general rule, anything below 0.20 is considered a poor question and in need of revision; items with a value between 0.20 and 0.30 are considered fair and could be improved upon, and items between 0.40 and 0.70 are considered good." (p. 167), and Haladyna, Rodriguez, and Stevens (2019) uses a boundary of 0.15 while stating "The application of 0.15 seems reasonable." (p. 358). None consider the number of items on the assessment nor the number of students used to calculate the potential error in a low PBC for a small set of students who took the assessment.

Measures From Item Response Theory

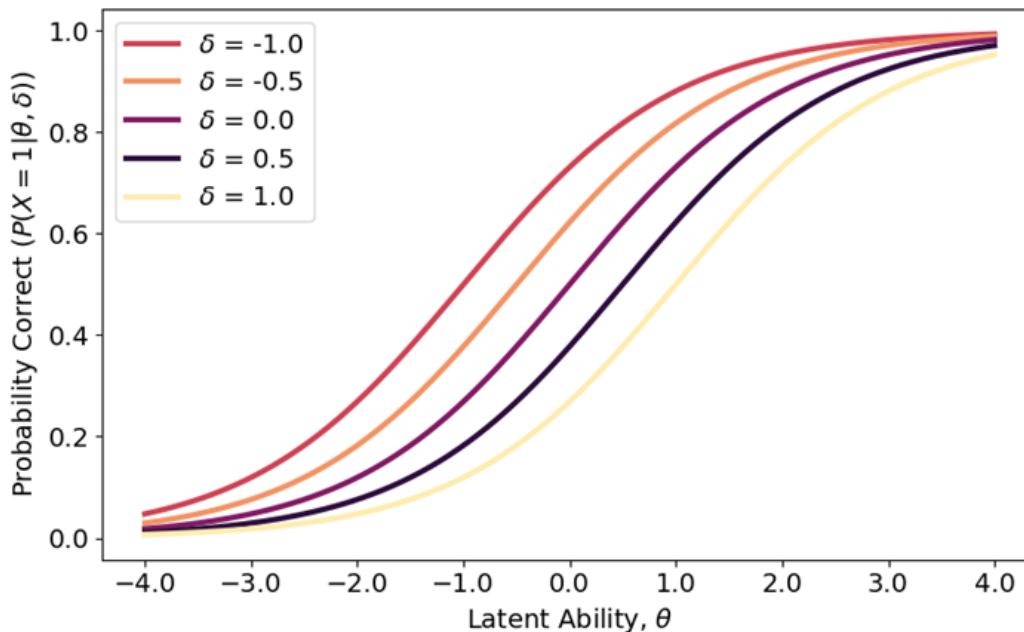
In contrast to Classical Test Theory, Item Response Theory (IRT) methodologies seek to learn a student's latent (unobservable) ability variable by analyzing the students' responses to each given item in the context of all students' responses to each given item. In general, this approach works by developing a mathematical model that relates a given item's estimated difficulty (based on all students' responses) to the learned latent ability variable (based on the individual student's responses to all items). The developed model can then be used to predict the probability that a given student (with corresponding latent ability) will correctly answer a given item (with corresponding estimated

difficulty). In this model, students with equal or higher latent ability than an item's estimated difficulty are predicted to correctly answer the item.

In the dichotomous setting, one can construct a one-parameter logistic (1PL) or three-parameter logistic (3PL) model that predicts that a student will answer an item correctly using a probability distribution function. When an item difficulty is fixed, one can plot the sigmoid curve relating ability to probability of answering the item correctly referred to as Item Characteristic Curve (ICC). **Figure 1** provides ICCs for items of difficulties (i.e. -1, -0.5, 0, 0.5, and 1) to illustrate how items with lower item difficulty reach higher correct probability more quickly. For example, a student with zero estimated ability would be expected to answer an item with -1 difficulty correctly about 73% of the time while the same student would be expected to correctly answer an item with 0.5 difficulty about 27% of the time.

Figure 1

Item Characteristic Curve for item difficulties (δ) of -1, -0.5, 0, 0.5, and 1



In such a logistic model, four initial assumptions must hold: (1) a unidimensional latent trait θ , (2) monotonically increasing latent trait, (3) item invariance, and (4) local independence of items. While there are methods for dealing with multiple latent traits (Reckase, 2009), one can start with the naive assumption that student mathematical ability is unidimensional monotonically increasing (e.g., as mathematical ability increases, score on assessment increases). Local independence of items is far less common in classroom assessment given the compounding nature of mathematics and similarity of topics between items tied to learning objectives. Item invariance assumes estimated item parameters are consistent across populations, which requires data from multiple populations to empirically test. The proposed methodology currently includes measures to test for local independence of items and will address how to assess data that fails local independence in future work.

Estimated Item Difficulty and Latent Ability

Two immediate consequences of training a one- or three-parameter model are “learning” an estimate for the item difficulty β for each assessment item and the latent ability θ for each student that took the exam. This means, for each item and student, there is a corresponding probability distribution function from which the probability that they will get an item correct. Beyond the utility in computing the probability a given student will answer an item correctly (which will provide the basis for analyses to come), these measures derived from training the model can provide insight into the student population and the items on the assessment. The distribution of student abilities and item difficulties can help inform the assessor of the spread of the students’ abilities and the items’ difficulty. For example, adaptive assessments use estimated student ability and item difficulty to provide students with questions that are challenging but not impossible (Baird et al., 2014). It is important to note that the latent ability θ and item difficulty β are unitless, and do not represent the same type of numerical relationship that PBC or item difficulty represents.

Infit and Outfit

After constructing a one- or three-parameter model of estimated student ability and item difficulty to predict the probability of answering an item correctly, one can consider how well the model fits the data. There are two residual-based fit statistics: outfit (outlier-sensitivity fit) and infit (information-weighted fit). Residuals, in the case of a logistic model, refers to the difference in the model’s calculated probability of a student answering an item correctly and the student’s observed score on the same item. The residual-based fit is calculated as this difference divided by the variance of the item response. Residual-based fit statistics can be measured against the standardized error of items as defined by one over the sum by items of the fit variance where outliers occur beyond $1 \pm SE_i$ (Wu et al., 2016). Since items with fit near one have the least discrimination power, it is suggested to use items with fit less than one. The outfit, or unweighted mean fit, is the average of the standardized fits without accounting for item variance. Outfit identifies patterns when item difficulty is far from student ability. For example, outfit might identify when students with high ability answer a low difficulty item incorrectly or vice versa. Poor outfit may suggest structural issues with the item, such as a typo. The infit, or weighted mean fit, is the average of the standardized fits while accounting for item variance. Infit identifies patterns when item difficulty is close to student ability. For example, infit might identify when students with ability near the item difficulty perform far better than expected by the model.

Measures from Distractor Analysis

Plausible, but incorrect, answers to the problem are referred to as distractors. Distractors can represent a wide variety of types of incorrect responses, and have been the topic of plenty of literature (Chamberlain Jr. & Jeter, 2019, 2020; Gierl et al., 2017). They can represent student misconceptions about the concept being assessed in an item or another concept; they can represent issues in manipulation or representation; they can even be created by manipulating the correct answer to the problem. Given the many types of thinking distractor answer choices can represent, they potentially play an integral role in the discriminative power of a multiple-choice item. For example, an item that has no distractors that represent plausible misconceptions could inadvertently direct students to the correct answer choice without any understanding of the underlying concept being evaluated.

Effective Distractors Frequency

An effective distractor is an incorrect answer choice that is chosen by students at a rate of at least 5% (Gierl et al., 2017). A distractor choice being chosen with some level of frequency indicates that it is likely to capture potential coherent thinking resulting from misconceptions about content. There are various opinions on how many effective distractors are optimal, with some authors arguing 3 total options, 2 distractors and a solution, is optimal (Haladyna et al., 2019) while others argue for as many distractors as are reasonable while ensuring each item has the same number of options throughout the assessment (Moreno et al., 2015). In light of these potential issues, the number of effective distractors an item has should hint to the discriminative power of the item itself.

Theoretical Framework

Utilizing item measures from Classic Test Theory, Item Response Theory, and Distractor Analysis, this section presents a summary of each metric that will inform labeling items as sufficiently discriminating for the purposes of an educational assessment. The proposed methodology begins by classifying each item-level metric into “ideal”, “acceptable”, or “poor” and analyzing how these categories collectively explain the discriminative power of the item.

Item Difficulty

As item difficulty is commonly used as one of many descriptive quantitative features, this framework utilizes the ideal item difficulties as suggested by Lord (1955) and standard deviation to create categories. Items within one standard deviation of 0.74 are considered to be “ideal” and items within two standard deviations as “acceptable.” Items outside of these ranges are “poor.”

Point-Biserial Correlation

To rigorously motivate the initial categorizations of PBCs, this paper considers two thresholds based on the number of items in the assessment (K) and number of students who answered items in the assessment (n).

The first threshold relates to the number of items in the assessment (K) and is based on the assumption of perfect determination. Given an exam with K items, the coefficient of determination for a single item is $r^2 = \frac{1}{K}$ and thus the correlation of any one item to the overall score would be $r = \frac{1}{\sqrt{K}}$.

Items with correlations at or above $\frac{1}{\sqrt{K}}$ suggest a reasonable explanation of the overall score based on the assessment size. For example, in an exam with 10 items, a reasonable correlation per item would be 0.32 while an exam with 20 items would have a reasonable correlation of 0.22 per item.

The second threshold relates to the number of students who answered items in the assessment (n) and is based on Fisher’s z-transformation (Fisher, 1915), which converts a correlation coefficient r into a normally distributed variable z . The standard error for Fisher’s z is $SE_z = \frac{1}{\sqrt{n-3}}$. By converting the correlations to a normally distributed variable, one can calculate a confidence interval with the formula $z' \pm z_{crit} * SE_z$. For significance 0.046, $z_{crit} = 2$. Thus when $z' < 2SE_z$, a confidence interval that contains 0 which would suggest the transformed correlation could be 0 for the assessment. Minimal

acceptance thresholds can avoid this scenario by ensuring the transformed correlation $z' \geq \frac{1}{\sqrt{n-3}}$, which means that $r \geq \tanh\left(\frac{2}{\sqrt{n-3}}\right)$. However, since $\tanh\left(\frac{2}{\sqrt{n-3}}\right) \approx \frac{2}{\sqrt{n-3}}$ when $n > 30$ and Fisher's z-transformation creates this bound using a normally distributed variable, the approximate simplification $r \geq \frac{2}{\sqrt{n-3}}$ is sufficient.

Now consider how these two bounds may interact for a varying number of items (K) and assessment sample size (n). The two bounds $\frac{1}{\sqrt{K}}$ and $\frac{2}{\sqrt{n-3}}$ are equal when $n = 4K + 3$, while $\frac{2}{\sqrt{n-3}} > \frac{1}{\sqrt{K}}$ when $n < 4K + 3$. This creates a natural separation when $n < 4K + 3$ of $\frac{1}{\sqrt{K}}$ to be a minimal acceptable PBC while $\frac{2}{\sqrt{n-3}}$ is the desired PBC. However, as sample size increases, $\frac{2}{\sqrt{n-3}}$ goes to 0 and would stop identifying minimally acceptable PBCs. The number of items on an assessment has a natural limit and so $\frac{1}{\sqrt{K}} - \frac{1}{\sqrt{n-3}}$ can be used as an estimate for minimally acceptable PBC, which will approach $\frac{1}{\sqrt{K}}$ as sample size increases. To ensure $\frac{1}{\sqrt{K}} - \frac{1}{\sqrt{n-3}}$ is greater than the critical bound of $\frac{2}{\sqrt{n-3}}$, restrict the $\frac{1}{\sqrt{K}} - \frac{1}{\sqrt{n-3}}$ to when $4K + 3 < n < 9K + 3$. A summary of the categorization of items based on the relationship between number of items (K) and assessment sample size (n) can be found in **Table 1**.

Table 1

Categorization of Point-Biserial Correlation (PBC)

$pb_{c_{acceptable}}$	$pb_{c_{ideal}}$	Condition	Example Classroom Circumstance
$\frac{1}{\sqrt{K}}$	$\frac{2}{\sqrt{n-3}}$	$n \leq 4K + 3$	Single class assessment E.g., $n = 32, K = 10$
$\frac{2}{\sqrt{n-3}}$	$\frac{1}{\sqrt{K}}$	$4K + 3 < n < 9K + 3$	Some sections, large assessment E.g., $n = 150, K = 20$
$\frac{1}{\sqrt{K}} - \frac{1}{\sqrt{n-3}}$	$\frac{1}{\sqrt{K}}$	$n \geq 9K + 3$	Many sections E.g., $n = 300, K = 20$

Note these categories are at least as strict as common static thresholds provided in the educational assessment literature while considering both sample size and assessment size. For example, Varma (2006) notes good items have point biserial correlation above 0.2. This 0.2 bound corresponds to either $K = 24$ or $n = 103$, which are common scenarios for either a final exam or a collection of sections for

a single course. The proposed PBC bounds also handle other regular cases of smaller classroom assessment such as $n = 32$ and $K = 10$, which would produce a minimum recommended PBC of 0.32 and an ideal PBC of 0.37.

Rasch Infit and Outfit

As noted in the literature review, residual-based fit outliers occur outside a standard error away from 1 (Wu et al., 2016). Since educational assessments commonly have numerous similar questions assessing the same Learning Objective, the model expects to see overfitting based on these similar questions. Thus, the methodology considers fits less than 1 minus standard error as “ideal”, fits at least less than 1 plus standard error as “acceptable”, and fits beyond 1 plus standard error as “poor.”

Effective Distractors

Ideally, every distractor would be “effective,” (each item would have 3 or 4 effective distractors, depending on whether the item has 4 or 5 options) meaning that it is chosen at least 5% of the time. Realistically, many factors contribute to the choice of a distractor on an item, such as the exam it is seen on (e.g., first exam versus final exam) and the level of students taking the exam. Thus this methodology considers an item “poor” if it has no effective distractors, “acceptable” if it has at least one effective distractor, and “ideal” if all distractors are effective distractors.

Summary of Classification Bounds by Measure

- **Item Difficulty** - Proportion of student sample who answered the item correctly. Ideally, item difficulty is normally distributed about 0.74 (for 4-option items) with as small a variance as possible. Items within one standard deviation of 0.74 are considered to be “ideal” and items within two standard deviations as “acceptable.” Items outside of these ranges are “poor.”
- **Point-Biserial Correlation** - Correlation between proportion of student sample who answered the item correctly and student sample overall score on assessment. Items at or above the maximum of $1/\sqrt{K}$ and $2/\sqrt{n-3}$ are considered to be “ideal.” Items above the minimum of $1/\sqrt{K}$ and $2/\sqrt{n-3}$ are considered “acceptable” while items below that minimum are considered “poor.” Table 1 summarizes the exact boundaries for ideal, acceptable, and poor based on the size of the assessment (K) and number of students taking the assessment (n).
- **Rasch Item Infit and Outfit** - Infit identifies patterns when estimated item difficulty is close to estimated student ability. Outfit identifies patterns when estimated item difficulty is far from estimated student ability. Fits less than 1 minus standard error are considered “ideal”, fits at least less than 1 plus standard error are considered “acceptable”, and fits beyond 1 plus standard error are considered “poor.”
- **Effective Distractors** - Distractors chosen at least 5% of the time. An item is considered “ideal” if all distractors are effective, “acceptable” if at least one distractor is effective, and “poor” if no distractor is effective.

Methods

To present a working example of how the framework for identifying measures can lead to classifying items, this paper presents results from four multiple-choice exams administered in College Algebra over the course of the Fall 2017 term at the University of Florida. The course consisted of 240 students divided into 9 sections. The study received an Exempt review via University of Florida IRB. **Table 2** presents a summary of the assessments.

Table 2

Summary of the Multiple-Choice (MC) Components of Exams Administered (Fall 2017)

		Exam 1		Exam 2			Exam 3			Exam 4		
		A	B	A	B	C	A	B	C	A	B	C
Number of MC Items	3 options			1	1	1						
	4 options	14	14	15	15	15	17	16	17	25	25	25
	5 options	2	2									
	total	16	16	16	16	16	17	16	17	25	25	25
Students that took the exam		117	123	75	76	77	75	72	73	76	75	74

Note: One item in Exam 3 Version B was thrown out due to typo.

The Python code to perform all of the analysis herein is available via the published GitHub repository with DOI 10.5281/zenodo.14782718.

Calculating Measures from Classical Test Theory

The CTT methods outlined in the theoretical framework and methodology require extracting every student to answer each exam item as well as the corresponding solution to each item. With this information, observed scores for each student on each assessment can be computed by dividing the number of items they answered correctly on an exam by the number of items on the exam. Similarly, item difficulty is computed by dividing the number of correct responses to an item by the number of students that attempted the item. With the observed score for each student and their responses to each item, point-biserial correlation can then be computed by evaluating Equation 1 for each item and computing the corresponding bounds based on the number of items on an exam and the number of students that took that exam.

Calculating Measures from Item Response Theory

Both a 1-parameter Rasch model (1PL) and 3-parameter logistic (3PL) model were constructed using the unconditional maximum likelihood estimation (UMLE) method. The steps for training the model are as follows:

1. Remove students and items with 100% scores and 0% scores

Step 2 computes the natural log of average scores (either by student or by item). Scores of 100% and 0% must be removed as $\log(0)$ and $\log(1/0)$ will produce errors.

2. Approximate ability and difficulty

1. Estimate ability per student using: $\theta_s = \ln \left(\frac{\delta_s}{1-\delta_s} \right)$, where $\delta_s = \frac{\sum q_{si}}{m}$ (e.g., sum of 1/0s for a single student divided by number of items student answered)
2. Estimate difficulty per item using: $b_i = \ln \left(\frac{1-\mu_i}{\mu_i} \right)$, where $\mu_i = \frac{\sum q_{si}}{n}$ (e.g., sum of 1/0s by all students for a single item divided by number of students who answered the item)

3. Standardize difficulty for mean 0

Adjust difficulty per item with: $\beta_i = b_i - \bar{b}$ (e.g., adjusted difficulty is estimated difficulty subtracted by average difficulty of all items)

4. Iterate until sum of squares of residuals provides minimal changes

1. **Calculate expected values:** probability of student answering item i correctly given a student's ability score and the item's difficulty $P(\theta_s, \beta_i)$. Note the Rasch 1PL model assumes $c_i = 0$ for all items.

$$P(\theta_s, \beta_i) = c_i + (1 - c_i) \frac{\exp(D(\theta_s - \beta_i))}{1 + \exp(D(\theta_s - \beta_i))}$$

2. **Re-calculate all θ_s and β_i** using the new expected values (steps 2 and 3).
3. **Calculate estimated variances of expected values:** $v_{si} = P(x_{si}) * (1 - P(x_{si}))$
4. **Calculate residuals between estimates and original data:** $e_{si} = q_{si} - P(x_{si})$
5. **Calculate sum of squares of residuals:**

$$e = \sum_i \sum_s (e_{si})^2$$

6. **If $e_k - e_{k+1} > 0.001$, re-calculate expected values and repeat:** Calculate revised θ_s and β_i and restart at step 4 a.

$$\theta_s = \theta_s + \frac{\sum_s e_{si}}{\sum_s v_{si}} \quad \beta_i = \beta_i - \frac{\sum_i e_{si}}{\sum_i v_{si}}$$

Calculating Measures from Distractor Analysis

The distractor-based methods outlined in the Methodology only require compiling a list of all student answers and a complete list of possible answer choices with their label of “distractor” or “solution.” Effective distractors per item only require direct computations of each distractor’s frequency of being chosen.

Item Discrimination

A summary dataset of all items can be constructed using the methods for generating item level measurements outlined in the previous “Methods” subsections. With this summary dataset, the “ideal,” “acceptable,” and “poor” thresholds can be directly computed, and each item directly assessed for its quality regarding each measure.

Results

An overview of each measurement described in the methodology with respect to the exams outlined in the Methods section follows. The underlying data for all Figures presented in the Results can be found in the Appendix.

Measures From Classical Test Theory

Overall, Classical Test Theory analysis paints a picture typical of multiple-choice assessments in lower division Mathematics courses: there is a tendency for a large percentage of the class to get nearly all the assessment items correct. This ultimately limits the amount of discriminative power of assessment items to determine specific conceptual misunderstandings.

Item difficulty

Figure 2A presents the distribution of item difficulty across the 11 exams in the study (data represented in Figure 2A can be found in Appendix Table 1). The distributions for item difficulty are not normal; they are mostly right skewed. This indicates that not only do most students perform well on all the exams, but most of the items are answered correctly by most students. **Figure 2B** presents the classification for items based on their difficulty (data represented in Figure 2B can be found in Appendix Table 2). Based on the bounds provided by the ideal 0.74 item difficulty and standard deviation, most items were marked as ideal with exception of Exam 2. For this exam, the more normal distribution of scores and lack of extreme outlier difficulties tightens the standard deviation and about one third of items for each version land in the acceptable range.

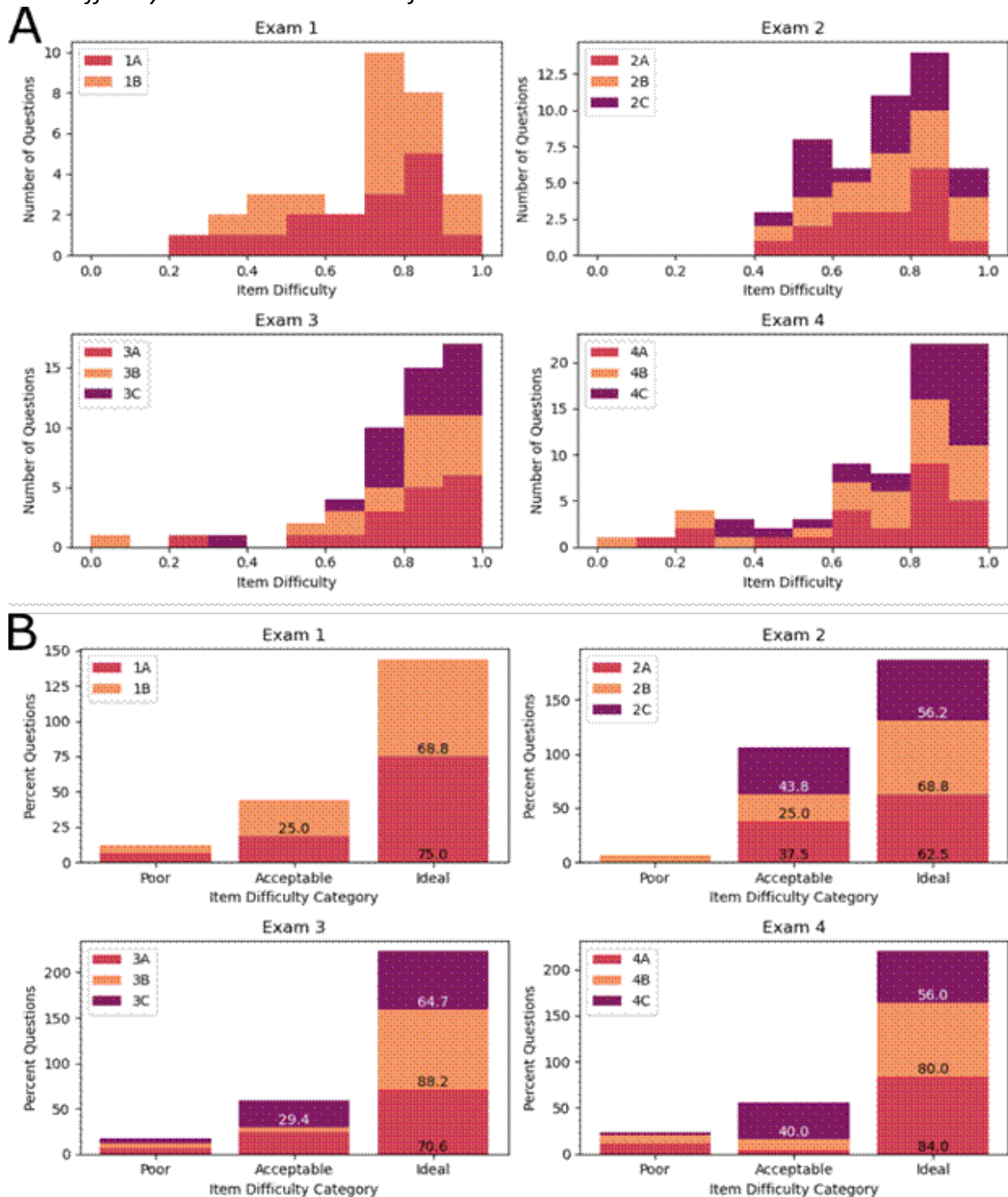
Figure 2*Item Difficulty Distribution and Classification*

Figure 2 Note: (A) Histograms showing the number of items for a given difficulty bin for each of the 11 exams. (B) Histograms of the percent of items in each category: Poor, Acceptable, Ideal based on the thresholds outlined in the “Methods” section.

Point-biserial correlation

The number of items in each category for each exam is presented in **Figure 3** (data represented in Figure 3 can be found in Appendix Table 3). Here is evidence that Exam 3 Version C is an anomaly: 62.5% of items presented poor point-biserial correlation. Otherwise, no exam has more than a few “poor” items in terms of point-biserial correlation, in fact, most items on most exams fall into the “ideal” category. This indicates that for most items, students’ performance on that item is reasonably indicative of their overall performance on the exam. This is consistent with the general tendency towards most students answering most items correctly.

Figure 3

Point-Biserial Correlation Classification

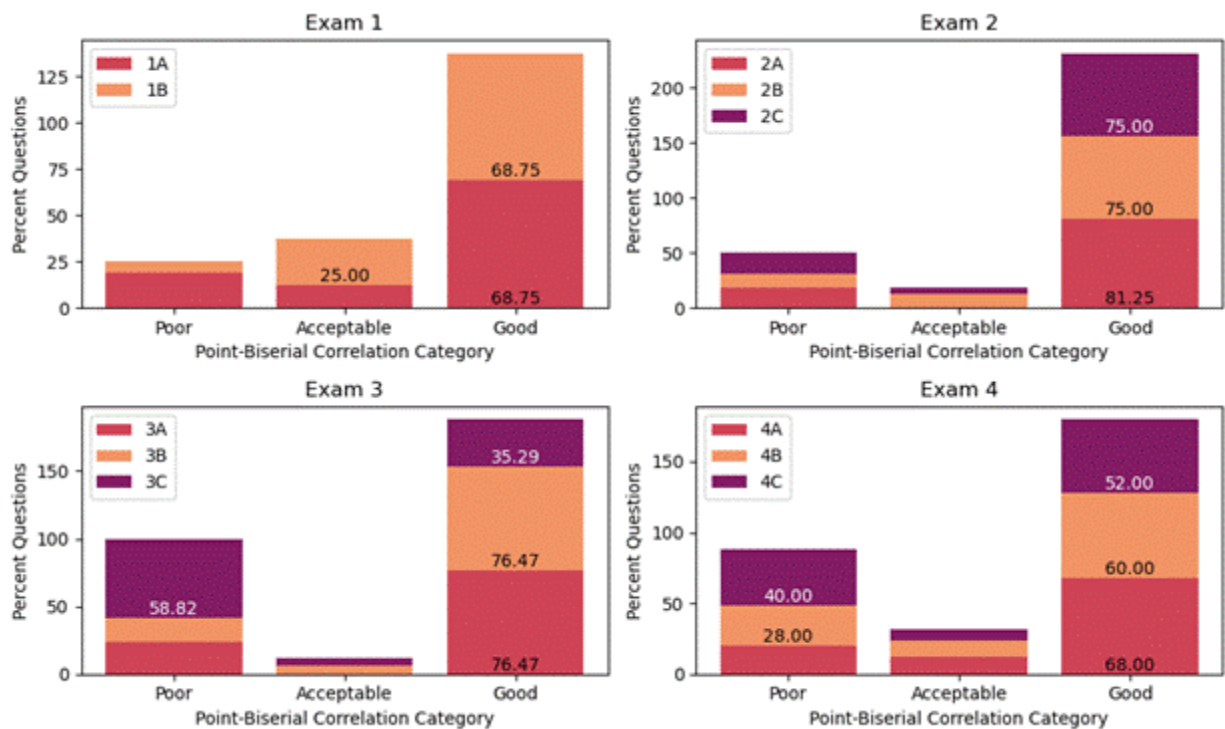


Figure 3 Note: Bar charts showing the percent of items on each exam that have a "poor," "acceptable," and "ideal" point-biserial correlation. Percentages for bars of the same color add up to 100%.

Measures From Item Response Theory

Overall, IRT results improve upon the ability to distinguish between item difficulty and student ability, moving distributions closer to normal. Few students and items were removed to perform the 1PL and 3PL models, also suggesting the removal of 100% items/students did not unduly affect the presented distributions. Moreover, exams 3B and 4B presented unusual estimated item difficulties and student abilities, suggesting unreliability with the version. Finally, exam 3A presented a high number of poor outfit items, suggesting either low-ability students were performing unusually well or high-ability

students were performing unusually poorly. A detailed look at each exam and each metric is presented below.

Estimated Item Difficulty

Figure 4 presents the 1PL and 3PL models for estimated item difficulty across exams 1, 2, 3, and 4 (data represented in Figure 4 can be found in Appendix Table 4-5). First, note the estimated item difficulty distributions are closer to normal than the item difficulty distributions for each exam. This represents an improvement in distinguishing between “hard” and “easy” items. Moreover, the distribution of estimated item difficulty for each exam is less uniform than the item difficulty distributions which reinforces the perception of moderate reliability between versions of the same exam. Note that the 1PL and 3PL models remove 0% and 100% scores on each item, which is removing data that may suggest that the true item difficulty distribution is more skewed than the estimated item difficulties present. Yet only 2 items were removed for producing scores of 100%: item 1 from Exam 3 Version C (3C01) and item 5 from Exam 4 Version B (4B05).

Exam 1 and Exam 2 estimated item difficulties for the 1PL (Rasch) and 3PL models are similar. Recall that the difference between these two models is the inclusion of a pseudo-guessing variable in the 3PL version that accounts for the inherent probability of guessing a multiple-choice item correctly. When including this variable, one item in 1A significantly increases in estimated item difficulty while in Exam 2 more items move closer to 0 estimated item difficulty.

In contrast, Exams 3 and 4 see a more significant change in estimated item difficulties between the 1PL and 3PL models. In both exams, version B items take strong shifts left once the model accounts for students’ ability to guess answers correctly. In these two Exams, the usefulness of including the 3PL model becomes clear as version B seems different from the other two versions.

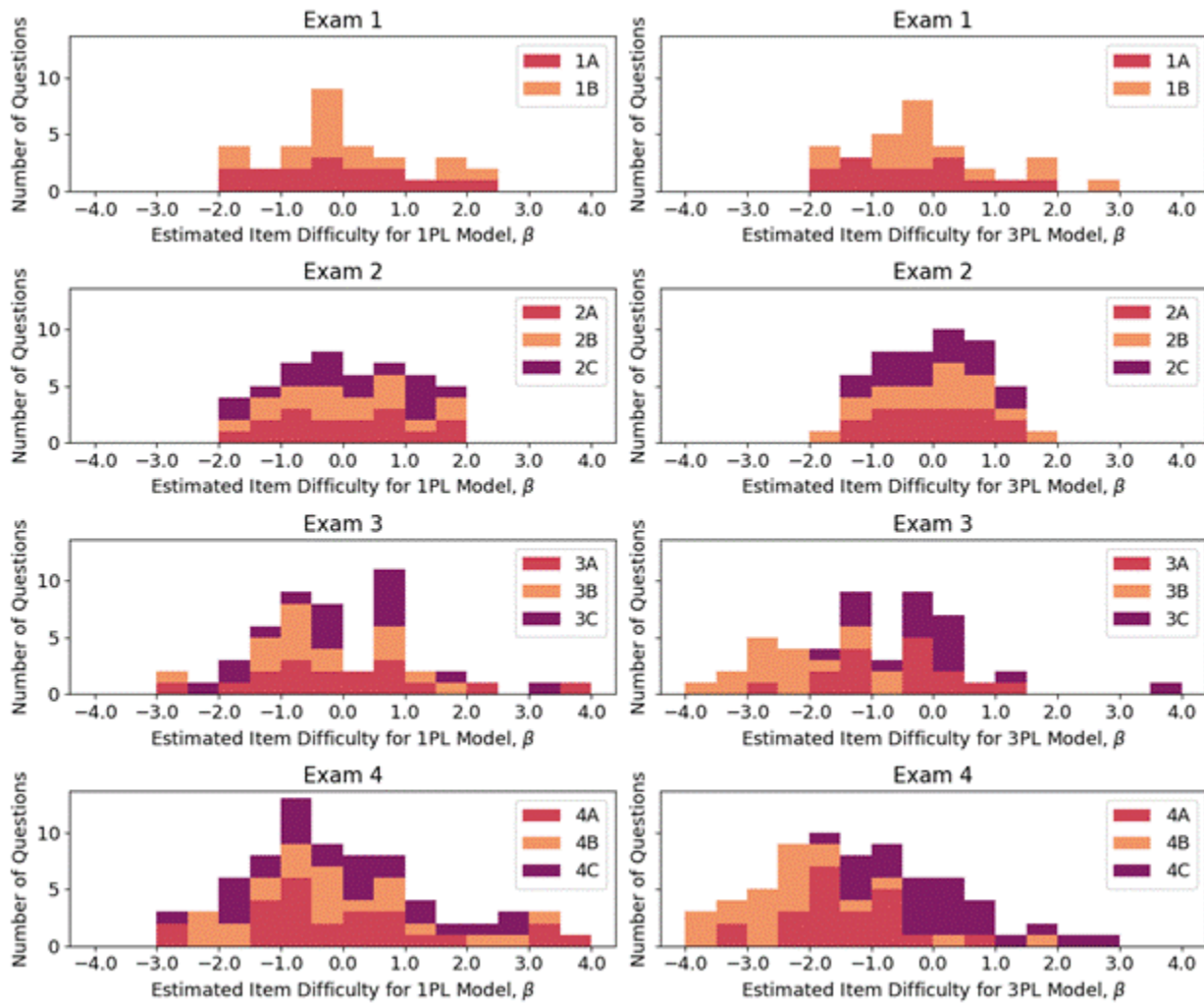
Figure 4*Estimated Item Difficulty Distributions*

Figure 4 Note: Distributions of the estimated item difficulty β for each exam learned from training the 1PL Rasch model (left columns) and the 3PL model (right columns) described in Equation 2.

Estimated Student Latent Ability

Figure 5 presents the estimated student latent ability across exams 1, 2, 3, and 4 (data represented in Figure 5 can be found in Appendix Table 6-7). Like the estimated item difficulty distributions, the estimated student latent ability distributions move toward a more normal distribution than the exam score distributions, allowing the models to more effectively distinguish between high-performing and low-performing students. Unlike the item difficulties, more students were removed from models due to 100% scores: 7 total in Exam 1, 10 total in Exam 2, 9 total in Exam 3, and 3 total in Exam 4. This represented the removal of 3%, 4%, 4%, and 1% of students from Exams 1, 2, 3, and 4, respectively.

As might be expected, the inclusion of the pseudo-guessing variable decreased the 3PL estimated student latent ability and thus very few students were modeled as more than 2. This suggests the 3PL model better estimates student latent ability across exams, where the estimated item difficulties were only sometimes different than their 1PL counterpart.

Like the identified differences with the 3B and 4B estimated item difficulties, the 3B and 4B 3PL estimated student latent abilities shifted out of sync with the 3A/C and 4A/C estimates. The entire 3B student estimated latent abilities are below 0 while nearly all 4B student estimated latent abilities are below 0. These are again patterns that cannot be detected from the exam score distribution and emphasize the usefulness of the 1PL and 3PL models when evaluating student performance.

Figure 5

Estimated Student Ability Distributions

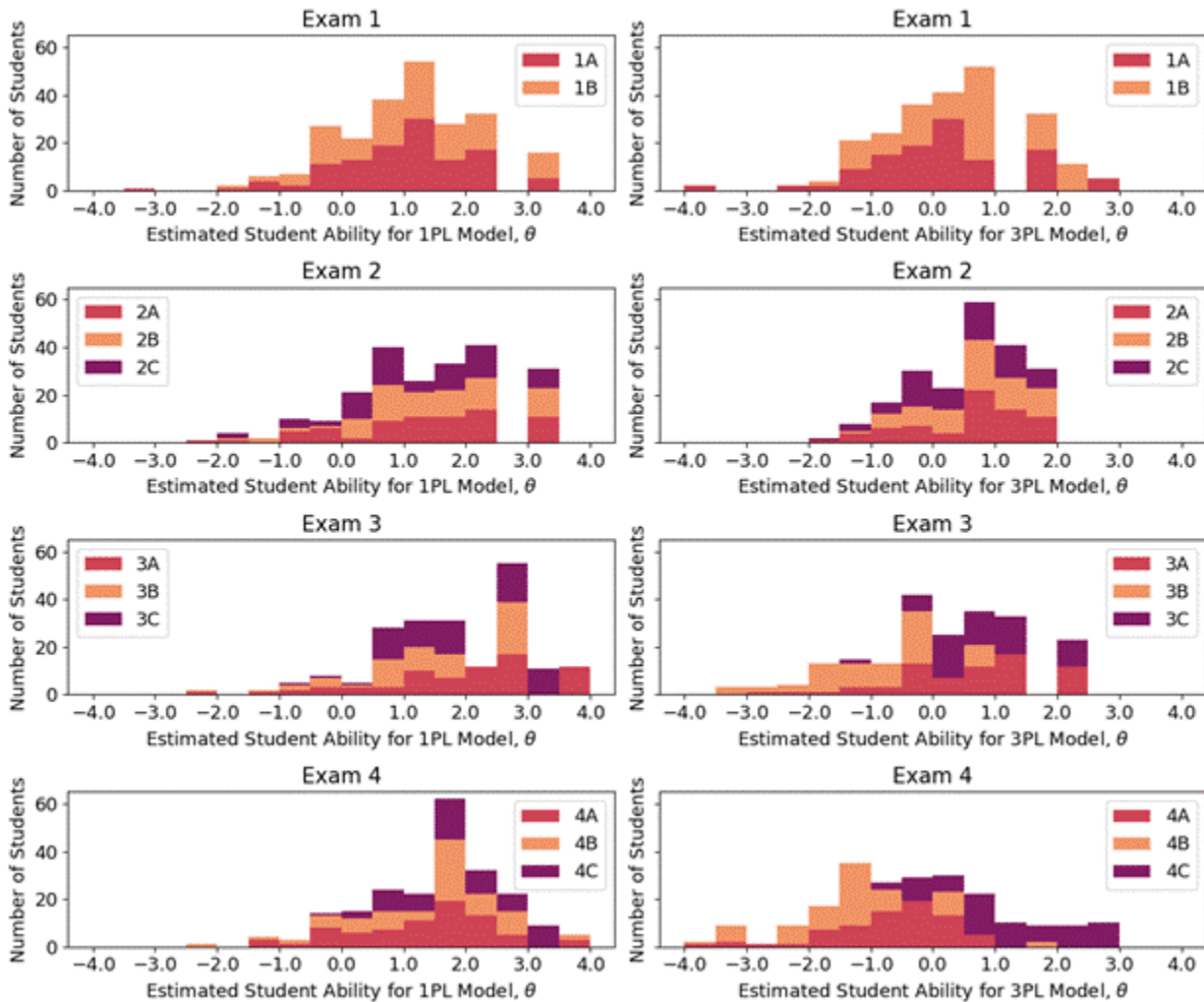


Figure 5 Note: Distributions of the estimated student ability θ for each exam learned from training the 1PL Rasch model (left columns) and the 3PL model (right columns) described in Equation 2.

Item Infit

Recall that Item Infit identifies patterns when item difficulty is close to student ability. Item Infit for each exam is presented in **Figure 6** and shows that both 1PL and 3PL models fit many of the exam items and land in the “Acceptable” range (data represented in Figure 6 can be found in Appendix Table 8-9). It is unlikely that infit alone will predict discrimination of any item, though a poor item infit may be particularly indicative of an item that needs to be considered for removal.

Figure 6

Infit Classifications

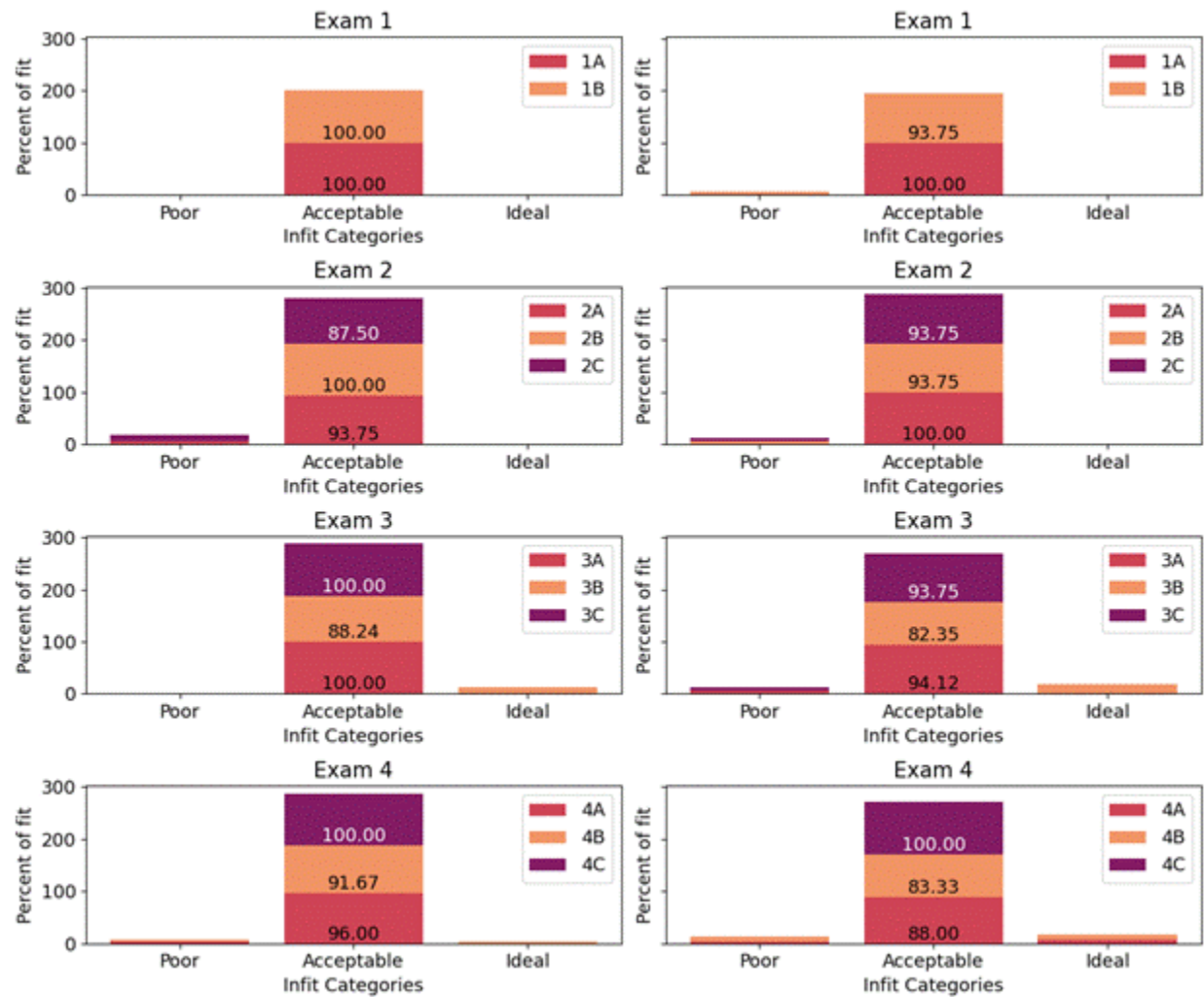


Figure 6 Note: Distributions of the item infit categories for each exam described in the “Infit” section of the “Methodology.” Left charts are item infits using the 1PL model while right charts are item infits using the 3PL model.

Item Outfit

Recall that outfit identifies patterns when item difficulty is far from student ability. Unlike Item Infit, **Figure 7** illustrates that all four exams show some ideal outfit (overfit outliers) and poor outfit (underfit outliers; data represented in Figure 7 can be found in Appendix Table 10-11). As expected with the increased frequency of item difficulties outside of -2 and 2 in exams 3 and 4, far more outfits of poor and ideal appeared in those exams. Of particular concern is the large percentage of poor outfit for Exam 3A. By considering the associated estimated item difficulty for each, instructors can understand whether this is low scoring students doing unusually well on the item or high scoring students doing unusually poorly on the item. In either case the item should be revised, but the way it may be revised would depend on which scenario is occurring.

Figure 7

Outfit Classifications

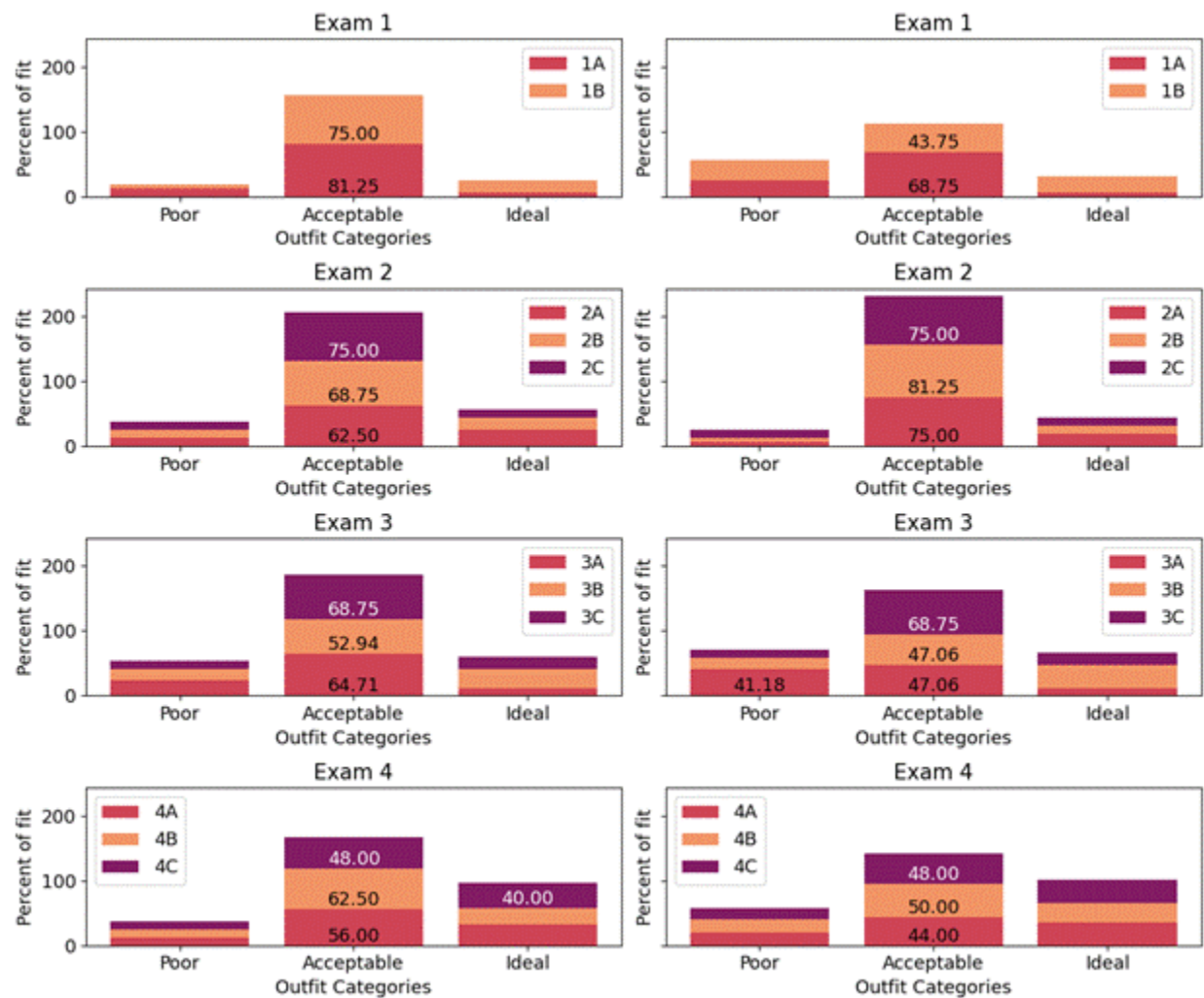


Figure 7 Note: Distributions of the outfit categories for each exam described in the “Outfit” section of the “Methodology.” Left charts are outfits using the 1PL Rasch model while right charts are outfits using the 3PL model.

Measures from Distractor Analysis

Presenting answer choices that effectively capture students’ misconceptions of the content mastery being evaluated can markedly improve the diagnostic utility of summative assessments. In turn, this targeted distractor creation methodology can improve the discriminative power of assessment items by providing an appropriate answer choice for students with specific misconceptions (leading them to select the distractor answer choice instead of potentially selecting at random). Similarly, poorly constructed distractor choices can increase the probability of selecting the correct answer choice for savvy students by allowing them to eliminate answer choices without any content mastery.

The results from analyzing the distractors in this study point to many distractors across the item set being selected less than 5% of the time. Subsequently, the percentage of items with more than one effective distractor (a distractor chosen more than 5% of the time) is around 50% for most of the exams. This implies that on most of the items, students can effectively reduce the number of likely answer choices to 2 (the solution and the effective distractor). This increase in probability of answering an item correctly at random (from 25% to 50% for most items) dramatically limits the discriminative power of these items.

With the cutoff of being selected at least 5% of the time for a distractor to be considered “effective,” **Figure 8** shows the percent of items with a given number of effective distractors for each exam (data represented in Figure 8 can be found in Appendix Table 12). There is a wide variety in the number of effective distractors per exam, likely due to the varying content and questions aligned to Learning Objectives. Exam 3 in particular appears to include many questions (approximately 33% per version) with no effective distractors. A more nuanced method for determining what is an effective distractor for high item difficulty questions may be needed in such circumstances as data from Exam 3.

Figure 8

Effective Distractor Classifications

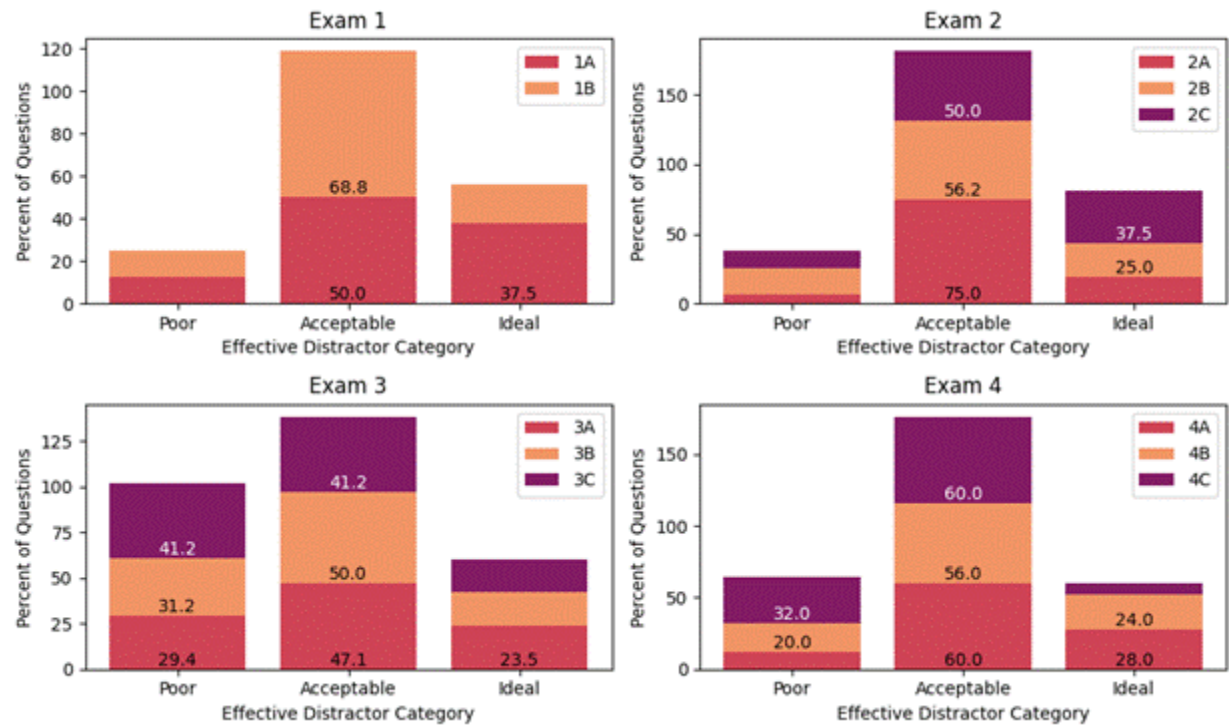


Figure 8 Note: Bar charts showing the percent of items on each exam that have a "poor," "acceptable," and "ideal" number of effective distractors. Percentages for bars of the same color add up to 100%.

Identifying Discriminative Items

Exam 4B is provided as an illustrative example as Exam 4 had the most questions (24) and version B had the most balanced classification of items. Combining all measurements for assessing assessment items with their corresponding "ideal," "acceptable," and "poor" categorizations results in the heat map shown in **Figure 9**. Each item in exam 4B is displayed with the five key measurements for identifying the discriminative power displayed on the x axis: 3PL item outfit, 3PL item infit, number of effective distractors, point-biserial correlation, and item difficulty. The cells for each item are color coded depending on whether the measurement falls into the "ideal," (green) "acceptable," (yellow) or "poor" (red) category.

The first takeaway from **Figure 9** is the usefulness of CTT metrics. Item 4B26 has a PBC of -0.11 and item difficulty of 0.09, which suggests that this item was so hard that almost no students were able to answer it correctly and that answering this question correctly had an inverse relationship with overall grade. This suggests that something about the item is drawing otherwise high-performing students to an unexpected response and needs to be investigated. This pairing of item difficulty and PBC also distinguishes between item 4B18 and 4B10. While both have similar item difficulty (0.23 and 0.28,

respectively), their PBC are far different (0.02 and 0.21, respectively) and would suggest 4B18 is in need of revision.

Items 4B07 and 4B20 are illustrative examples for the efficacy of the integrated methodology. Item 4B07 presents ideal item difficulty and PBC (0.92 and 0.45, respectively), however shows poor item outfit of 5.82. Such a high item outfit suggests unpredictability with respect to students' estimated ability and the low difficulty of the item, prompting an instructor to investigate why high-performing students would get this low difficulty item incorrect.

The usefulness of including effective distractors appears to be relative to the item difficulty. Consider items 4B19 and 4B24: both had relatively equal item difficulty (0.67 and 0.69, respectively) and different numbers of effective distractors (1 and 3, respectively). This potential difference is emphasized by the item outfit for each (1.44 and 0.44, respectively) which suggests that more effective distractors point to a more stable relationship between student ability and item difficulty. It may be necessary to create further metrics related to effective distractors though as it is unclear how choosing these anticipated but incorrect answers impacts the predictive models.

Figure 9

Classification of Measures for Exam 4 Version B

	Item Outfit	Item Infit	Effective Distractors	PBC	Item Difficulty
4B02 -	0.93	0.97	3.00	0.18	0.39
4B03 -	1.01	1.05	3.00	0.17	0.53
4B04 -	0.55	0.81	1.00	0.53	0.88
4B06 -	0.32	0.74	0.00	0.49	0.95
4B07 -	5.82	0.85	0.00	0.45	0.92
4B08 -	0.55	0.90	1.00	0.49	0.89
4B09 -	1.66	1.42	0.00	0.18	0.96
4B10 -	0.97	1.00	3.00	0.21	0.28
4B11 -	0.32	1.04	1.00	0.42	0.93
4B12 -	0.87	0.98	3.00	0.43	0.77
4B13 -	1.56	1.18	2.00	0.21	0.65
4B14 -	0.81	1.10	2.00	0.37	0.88
4B15 -	1.07	1.17	1.00	0.25	0.80
4B16 -	0.93	1.02	1.00	0.40	0.89
4B17 -	0.89	1.05	2.00	0.36	0.76
4B18 -	0.90	0.90	2.00	0.02	0.23
4B19 -	1.44	1.32	1.00	0.05	0.67
4B20 -	2.73	1.59	0.00	0.12	0.96
4B21 -	0.55	0.74	2.00	0.58	0.88
4B22 -	0.79	0.88	2.00	0.49	0.80
4B23 -	0.73	0.94	2.00	0.46	0.83
4B24 -	0.44	0.58	3.00	0.72	0.69
4B25 -	0.74	1.06	2.00	0.34	0.77
4B26 -	0.58	0.58	3.00	-0.11	0.09

Figure 9 Note: Heatmap showing the item summary for exam 4B. Green items achieve the “ideal” thresholds outlined in the Methodology section, yellow items achieve the “acceptable” thresholds, and red items are “poor.”

Discussion

While the literature commonly presents item difficulty as either similar or different from the norm (e.g., Rush, Rakin, & White, 2016), this study investigated classification bounds based on psychometrics of predictive items as originally suggested by Lord (1955). As typical for educational assessment data (Ho & Yu, 2015), the methodology was applied to non-normal educational assessment data. While the bounds distinguished between acceptable and ideal items for one of the four provided exams, it otherwise filtered out very few items to be classified as “Poor.” This provides information to instructors looking to ensure their difficulties do not deviate too much and can be useful for investigating potentially problematic items, such as those with a typo that preclude students from answering as expected. More work will be needed to investigate if the use of 1 and 2 standard deviations away is useful for predicting item classification.

The dynamic bounds for PBC classification appear to identify poorly-performing items while attending to the size of the assessment as well as number of students taking said assessment. These bounds also generally align with the static bounds suggested in the literature (McGhee & Ball, 2009; Varma, 2006). A classification system using just PBC would not be ideal as the higher the item difficulty for the item, the closer to 0 the PBC for the item will be. High item difficulty and low PBC items may be desired for critical learning objectives students are expected to show mastery on (McGhee & Ball, 2009) while low item difficulty and low PBC items may suggest random guessing by students and need to be revised.

Item fits (infit and outfit) were effective at distinguishing between items with ideal PBC by first modeling estimated student ability and item difficulty to determine how well student responses could be predicted by the model. Outliers on the model identified items that were in need of revision. The decision to classify fit into three categories by fit outliers did not seem useful for the presented dataset though and so a competing classification with only two categories (“acceptable” and “poor”) will be used to determine the usefulness of the “ideal” threshold for item fits in particular.

Effective distractor classification added a nuanced consideration to the CTT and IRT metrics. Zero effective distractors necessarily occurred with high item difficulty and some effective distractors naturally occurred with low item difficulty. While some authors suggest that the number of options should be the same for all items on an educational assessment (Moreno Martínez, & Muñiz, 2015)), nearly 20% of questions per exam had 3 effective distractors and showed the potential for 4-option questions. Future work will incorporate a sliding scale to identify low-frequency distractors from high item difficulty questions as presented in Raymond, Stevens, and Bucak (2019).

Conclusion

This paper presented an integrated methodology that combines measures from classical test theory, item response theory, and distractor analysis to identify discriminative items in multiple choice mathematics assessments. The methodology either adopts current suggestions for thresholds or provides explicit, statistically motivated thresholds for each measurement so that each item could be procedurally categorized as “ideal,” “acceptable,” and “poor” with respect to each measurement. This methodology provides instructors and researchers with a high-level view of each item in an assessment that can be easily interpreted to see the relationships between each measurement and suggest items

that are effective and ineffective at discriminating between high and low performing students. This allows the instructor to quickly evaluate items for retention and removal for future exams. Future refinement of this methodology will include providing specific algorithms for relating how the categorical measures relate to one another with the goal of a simple recommendation to either remove, revise, or keep each item.

Beyond the value of an instructor deciding which items are effective and should be retained, the items highlighted by this integrated methodology present the opportunity to determine why certain items are more effective than others and should be qualitatively analyzed by researchers. Qualitative analysis of distractors identified using this integrated methodology remains to be studied and will be the subject of future work. Moreover, machine learning classification algorithms may be performed based on the goal of an assessment to rigorously classify assessments.

Acknowledgements

Darryl Chamberlain Jr. acknowledges the support of NSF award number 2315560.

Appendix: Data for Presented Figures

Table 1

Data Represented in Figure 2A

Bin Range	1A	1B	2A	2B	2C	3A	3B	3C	4A	4B	4C
0 - 0.1	0	0	0	0	0	0	1	0	0	1	0
0.1 - 0.2	0	0	0	0	0	0	0	0	1	0	0
0.2 - 0.3	1	0	0	0	0	1	0	0	2	2	0
0.3 - 0.4	1	1	0	0	0	0	0	1	0	1	2
0.4 - 0.5	1	2	1	1	1	0	0	0	1	0	1
0.5 - 0.6	2	1	2	2	4	1	1	0	1	1	1
0.6 - 0.7	2	0	3	2	1	1	2	1	4	3	2
0.7 - 0.8	3	7	3	4	4	3	2	5	2	4	2
0.8 - 0.9	5	3	6	4	4	5	6	4	9	7	6
0.9 - 1	1	2	1	3	2	6	5	6	5	6	11

Table 1 Note: Data used for the histograms showing the number of items in a given difficulty range for each of the 11 exams.

Table 2

Data Represented in Figure 2B

Category	1A	1B	2A	2B	2C	3A	3B	3C	4A	4B	4C
Poor	1	1	0	1	0	1	1	1	3	2	1
Acceptable	3	4	6	4	7	4	1	5	1	3	10
Ideal	12	11	10	11	9	12	15	11	21	20	14

Table 2 Note: Data used for the bar charts showing the number of items for a given difficulty category for each of the 11 exams.

Table 3

Data Represented in Figure 3

Category	1A	1B	2A	2B	2C	3A	3B	3C	4A	4B	4C
Poor	3	1	3	2	3	4	3	10	5	7	10
Acceptable	2	4	0	2	1	0	1	1	3	3	2
Ideal	11	11	13	12	12	13	13	6	17	15	13

Table 3 Note: Data used for bar charts showing the percent of items on each exam that have a "poor," "acceptable," and "ideal" point-biserial correlation.

Table 4*Data Represented in Figure 4, Left Column*

Bin Range	1A	1B	2A	2B	2C	3A	3B	3C	4A	4B	4C
-4 - -3.5	0	0	0	0	0	0	0	0	0	0	0
-3.5 - -3	0	0	0	0	0	0	0	0	0	0	0
-3 - -2.5	0	0	0	0	0	1	1	0	2	0	1
-2.5 - -2	0	0	0	0	0	0	0	1	0	3	0
-2 - -1.5	2	2	1	1	2	1	0	2	0	2	4
-1.5 - -1	2	0	2	2	1	2	3	1	4	2	2
-1 - -0.5	2	2	3	2	2	3	5	1	6	3	4
-0.5 - 0	3	6	2	3	3	2	2	4	2	5	2
0 - 0.5	2	2	2	2	2	2	0	0	3	1	4
0.5 - 1	2	1	3	3	1	3	3	5	3	3	2
1 - 1.5	1	0	1	1	4	1	1	0	1	1	2
1.5 - 2	1	2	2	2	1	0	1	1	1	0	1
2 - 2.5	1	1	0	0	0	1	0	0	0	1	1
2.5 - 3	0	0	0	0	0	0	0	0	0	1	2
3 - 3.5	0	0	0	0	0	0	0	1	2	1	0
3.5 - 4	0	0	0	0	0	1	0	0	1	0	0

Table 4 Note: Data for distributions of the estimated item difficulty β for each exam learned from training the 1PL Rasch model.

Table 5*Data Represented in Figure 4, Right Column*

Bin Range	1A	1B	2A	2B	2C	3A	3B	3C	4A	4B	4C
-4 - -3.5	0	0	0	0	0	0	1	0	0	3	0
-3.5 - -3	0	0	0	0	0	0	2	0	2	2	0
-3 - -2.5	0	0	0	0	0	1	4	0	0	5	0
-2.5 - -2	0	0	0	0	0	0	4	0	3	6	0
-2 - -1.5	2	2	0	1	0	2	1	1	7	2	1
-1.5 - -1	3	0	2	2	2	4	2	3	3	1	4
-1 - -0.5	2	3	3	2	3	0	2	1	5	1	3
-0.5 - 0	2	6	3	2	3	5	0	4	1	0	5
0 - 0.5	3	1	3	4	3	2	0	5	0	1	5
0.5 - 1	1	1	3	3	3	1	0	0	1	0	3
1 - 1.5	1	0	2	1	2	1	0	1	0	0	1
1.5 - 2	1	2	0	1	0	0	0	0	0	1	1
2 - 2.5	0	0	0	0	0	0	0	0	0	0	1
2.5 - 3	0	1	0	0	0	0	0	0	0	0	1
3 - 3.5	0	0	0	0	0	0	0	0	0	0	0
3.5 - 4	0	0	0	0	0	0	0	1	0	0	0

Table 5 Note: Data for distributions of the estimated item difficulty β for each exam learned from training the 3PL model.

Table 6*Data Represented in Figure 5, Left Column*

Bin Range	1A	1B	2A	2B	2C	3A	3B	3C	4A	4B	4C
-4 - -3.5	0	0	0	0	0	0	0	0	0	0	0
-3.5 - -3	1	0	0	0	0	0	0	0	0	0	0
-3 - -2.5	0	0	0	0	0	0	0	0	0	0	0
-2.5 - -2	0	0	1	0	0	1	1	0	0	1	0
-2 - -1.5	1	1	1	1	2	0	0	0	0	0	0
-1.5 - -1	4	2	0	2	0	1	1	0	3	1	0
-1 - -0.5	2	5	5	1	4	2	2	1	1	2	0
-0.5 - 0	11	16	6	1	2	3	4	1	8	5	1
0 - 0.5	13	9	2	8	11	3	1	1	6	6	3
0.5 - 1	19	19	9	15	16	3	12	13	7	8	9
1 - 1.5	30	24	11	10	5	10	10	11	11	4	7

1.5 - 2	13	15	11	11	11	7	10	14	19	26	17
2 - 2.5	17	15	14	13	14	12	0	0	13	9	10
2.5 - 3	0	0	0	0	0	17	22	16	5	10	7
3 - 3.5	5	11	11	12	8	0	0	11	0	0	9
3.5 - 4	0	0	0	0	0	12	0	0	3	2	0

Table 6 Note: Data for distributions of the estimated student ability θ for each exam learned from training the 1PL Rasch model.

Table 7

Data Represented in Figure 5, Right Column

Bin Range	1A	1B	2A	2B	2C	3A	3B	3C	4A	4B	4C
-4 - -3.5	2	0	0	0	0	0	0	0	1	1	0
-3.5 - -3	0	0	0	0	0	0	3	0	2	7	0
-3 - -2.5	0	0	0	0	0	1	2	0	1	0	0
-2.5 - -2	2	0	0	0	0	1	3	0	1	8	0
-2 - -1.5	2	2	1	0	1	1	12	0	7	10	0
-1.5 - -1	9	12	4	1	3	3	10	2	9	26	0
-1 - -0.5	15	9	6	6	5	3	10	0	15	9	3
-0.5 - 0	19	17	7	8	15	13	22	7	19	0	10
0 - 0.5	30	11	4	10	9	7	0	18	13	10	7
0.5 - 1	13	39	22	21	16	12	9	14	5	0	17
1 - 1.5	0	0	14	13	14	17	0	16	0	0	10
1.5 - 2	17	15	11	12	8	0	0	0	0	2	7
2 - 2.5	0	11	0	0	0	12	0	11	0	0	9
2.5 - 3	5	0	0	0	0	0	0	0	0	0	10
3 - 3.5	0	0	0	0	0	0	0	0	0	0	0
3.5 - 4	0	0	0	0	0	0	0	0	0	0	0

Table 7 Note: Data associated with Figure 5 distributions of the estimated student ability θ for each exam learned from training the 3PL model.

Table 8*Data Represented in Figure 6, Left Column*

Category	1A	1B	2A	2B	2C	3A	3B	3C	4A	4B	4C
Poor	0	0	1	0	2	0	0	0	1	1	0
Acceptable	16	16	15	16	14	17	15	16	24	22	25
Ideal	0	0	0	0	0	0	2	0	0	1	0

Table 8 Note: Data for distributions of the item infits for each exam from training the 1PL model.

Table 9*Data Represented in Figure 6, Right Column*

Category	1A	1B	2A	2B	2C	3A	3B	3C	4A	4B	4C
Poor	0	1	0	1	1	1	0	1	1	2	0
Acceptable	16	15	16	15	15	16	14	15	22	20	25
Ideal	0	0	0	0	0	0	3	0	2	2	0

Table 9 Note: Data for distributions of the item infits for each exam from training the 3PL model.

Table 10*Data Represented in Figure 7, Left Column*

Category	1A	1B	2A	2B	2C	3A	3B	3C	4A	4B	4C
Poor	2	1	2	2	2	4	3	2	3	3	3
Acceptable	13	12	10	11	12	11	9	11	14	15	12
Ideal	1	3	4	3	2	2	5	3	8	6	10

Table 10 Note. Data for distributions of the item outfit for each exam from training the 1PL model.

Table 11*Data Represented in Figure 7, Right Column*

Category	1A	1B	2A	2B	2C	3A	3B	3C	4A	4B	4C
Poor	4	5	1	1	2	7	3	2	5	5	4
Acceptable	11	7	12	13	12	8	8	11	11	12	12
Ideal	1	4	3	2	2	2	6	3	9	7	9

Table 11 Note: Data for distributions of the item outfit for each exam from training the 3PL model.

Table 12*Data Represented in Figure 8*

Effective Distractors	1A	1B	2A	2B	2C	3A	3B	3C	4A	4B	4C
Poor	12.50	12.50	6.25	18.75	12.50	29.41	29.41	41.18	12.00	20.00	32.00
Acceptable	50.00	68.75	81.25	62.50	56.25	47.06	52.94	41.18	60.00	56.00	60.00
Ideal	37.50	18.75	12.50	18.75	31.25	23.53	17.65	17.65	28.00	24.00	8.00

Table 12 Note: Data for percent of items on each exam that have a "poor," "acceptable," and "ideal" number of effective distractors.

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